Sahaj Bhadja

(213) 284-4651 bhadja@usc.edu sahaj-bhadja sahajbhadja sahaj-bhadja

Summary

Software engineer pursuing an MSCS at USC with 2+ years building cross-platform software, APIs, test automation, and cloud-backed systems. Strong CS fundamentals across operating systems, data structures, OOP, Linux, C/C++, Java, Python, Git, debugging, CI/CD, and performance-minded engineering.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Built SQL-backed platform workflows for **500+** users, translating requirements into schemas, APIs, validation logic, and release tasks for reliable cloud-hosted services.
- Improved runtime performance by optimizing indexed access paths and background jobs, reducing dashboard latency from **1.2s to 350ms** while strengthening deployment and troubleshooting handoffs.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Developed Python and JavaScript services handling **10,000+** daily requests, implementing REST APIs, concurrent file processing, and model-backed workflows with production observability.
- Debugged OCR, database, file-ingestion, and LLM execution failures, documenting root causes and fixes that reduced instructor feedback turnaround by **60%**.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Created Python evaluation pipelines over **100,000+** records, producing reproducible metrics, logs, and debugging artifacts for model behavior analysis and data-quality validation.
- Collaborated with research and engineering stakeholders to translate experiments into maintainable scripts, reports, and issue summaries for iterative AI/ML software development.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Engineered cross-platform microservices and ETL workflows using Python, OOP patterns, AWS Glue, and S3, improving API response time by **30%**.
- Maintained Jenkins pipelines and **20+** PyTest/Mockito suites across Windows, Linux, and macOS validation flows, increasing CI reliability by **68%**.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Built modular services with REST APIs, WebSockets, and SQL/NoSQL persistence, supporting **2M+** synchronized records across distributed product workflows.
- Automated Docker and Jenkins build/test flows, cutting release time by **50%** and improving repeatable deployments, debugging, and documentation for engineering teams.

Skills

Systems & OS: Operating systems, Linux/Unix, command line, process scheduling, concurrency, memory management, networking basics
Languages: Python, Java, JavaScript, TypeScript, C, C++, SQL, Bash, HTML/CSS
Platform Tools: Git, Docker, Jenkins, GitHub Actions, Linux debugging, logs, Android Studio/ADB familiarity, Make familiarity
Backend & AI: REST APIs, Flask, FastAPI, Node.js, PostgreSQL, MySQL, MongoDB, AWS S3/Glue/EC2, PyTorch/Transformers

Projects

CPU Scheduling Simulator Live | *C++, React.js, JavaScript*
- Implemented FCFS, SJF, SRT, RR, priority, and MLQ schedulers with metric calculation and Gantt visualization across **1K+** process simulations.

Chat2SQL Code | *Java, TypeScript, SQL, Docker*
- Built an AI-powered SQL workspace supporting **10+** databases, working across Java services, TypeScript UI, Docker setup, schema exploration, and query execution.

Linguistix Code | *Python, AWS, Docker, CI/CD*
- Developed a containerized REST data-processing API with AWS deployment, CI/CD releases, validation checks, and reproducible execution, achieving **87%** classification accuracy.